

Unequal Job Opportunities and Well-Being Inequality

A Latent-Jobs Structural Decomposition

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Thank you, and thank you to Sebastian for discussing the paper. This talk asks how much of the inequality in money-metric well-being is associated with unequal job opportunities rather than with differences in what people prefer. I build the answer in four steps: a model of labour supply as choice among latent jobs, which separates preferences from opportunities; two ways of turning choices into euros of well-being; counterfactuals that equalise one source of heterogeneity at a time; and a Shapley rule that allocates the change. The punchline is that the answer depends on the welfare question: prospects or attained outcomes.

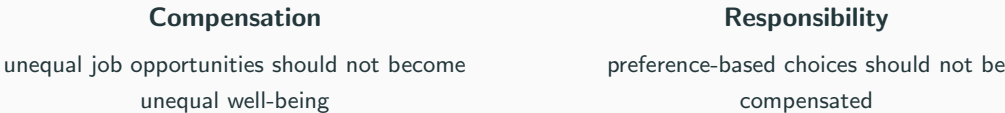
Observed	Well-being
C_i	M_i
Gini of consumption	Gini of the attained-bundle money metric
single adults 0.2633 couples 0.2268	single adults 0.2498 couples 0.1974
low earnings $\Leftarrow$ tastes or opportunities or resources	
equivalised, household-weighted, within each population	

Two people can work the same hours for the same hourly pay and be very differently placed. One chose that job from several that were open; the other took the only offer available. Their incomes are identical and their circumstances are not.

Low earnings can describe someone who values leisure and chose short hours, someone who cannot obtain a well-paid job, or someone whose household resources make a different arrangement affordable. An income distribution records outcomes with these explanations already mixed in.

Once the leisure a job costs is valued, the dispersion of well-being is not the dispersion of consumption. Single-adult and couple households are separate populations, never compared with each other.

The conflict



Compensation vs. Responsibility

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet]

The conflict that makes this paper necessary.

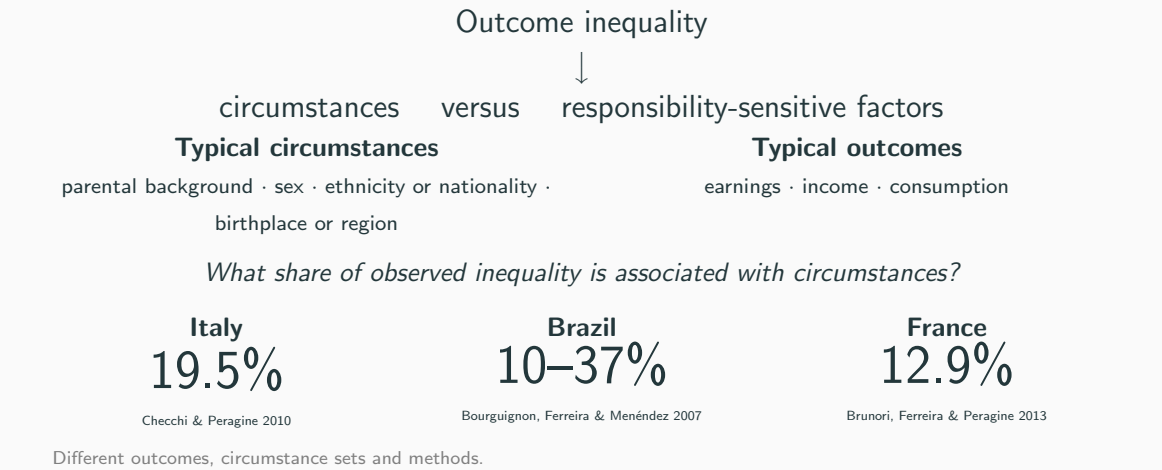
A fair well-being comparison would like to respect two principles. Compensation: people should not end up worse off because they face worse job opportunities. Responsibility: people are responsible for their preferences, so a choice that reflects taste does not generate a compensation claim.

Each principle is attractive on its own, but no well-being measure satisfies both in full. The literature, following Fleurbaey and Maniquet, is organised around compromises. To know what to compensate, we must separate what people prefer from what they can reach, and observed choices mix the two.

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

The whole talk on one slide. The first subquestion is behavioural: can we read preferences off choices, or do opportunities get in the way? The second is normative: does it matter whether we value the prospect a household faces or the outcome it attains? The third is the accounting: with preferences, local access and earning opportunities as the three channels, how is inequality allocated?



This is the empirical inequality-of-opportunity question. Take an outcome distribution, split its determinants into circumstances, for which people are not responsible, and responsibility-sensitive factors, and ask what share of observed inequality is associated with circumstances.

The benchmarks are well known. In Italy, parental education accounts for about one fifth of gross-earnings inequality in Checchi and Peragine. In urban Brazil, five observed circumstances account for between ten and thirty-seven per cent of male earnings inequality across cohorts in Bourguignon, Ferreira and Menéndez. For France, the cross-country compilation of Brunori, Ferreira and Peragine reports about thirteen per cent for post-tax individual earnings.

These numbers use different outcomes, circumstance sets and methods, so they are orientation, not a league table. They are lower-bound shares of earnings or income inequality attributed to observed circumstances. The share in this paper is a share of welfare inequality associated with estimated latent opportunity components, with household resources and needs held fixed. The backup slide has the full citations.

From circumstances to job opportunities

	Empirical EOp	This paper
Opportunity	observed circumstances or types	estimated latent-job distribution
Outcome	earnings, income, consumption	money-metric well-being
Structural objects	—	preferences · access · earning opportunities
Strategy	compare circumstance-defined distributions	equalise structural components, recompute welfare, decompose inequality

An inequality-of-opportunity question with a structural opportunity object.

The same question, with a different opportunity object. Empirical inequality of opportunity defines opportunity by observed circumstances or types and compares outcome distributions across them. Here, opportunity is the estimated distribution of latent jobs a household can reach, the outcome is money-metric well-being, and the structural objects are preferences, access and earning opportunities. The strategy is to equalise those components inside the model, recompute welfare for every household and decompose the change in inequality. That is an inequality-of-opportunity question with a structural opportunity object.

Latent jobs

Dagsvik & Jia
Aaberge–Colombino tradition
Capéau, Decoster & Dekkers

Welfare

Fleurbaey & Maniquet
Decoster & Haan
Jacquet, Jia & Thoresen

Decomposition

Shorrocks · Shapley

The contribution is the structural equalisation of estimated preferences, access and earning opportunities followed by recomputation of money-metric welfare inequality.

Three literatures supply the ingredients. Labour supply as choice among latent jobs, from Dagsvik and Jia and the Aaberge–Colombino tradition, set out for applied work by Capéau, Decoster and Dekkers. Money-metric welfare with heterogeneous preferences, from Fleurbaey and Maniquet, Decoster and Haan, and Jacquet, Jia and Thoresen. And the Shapley decomposition of Shorrocks.

The contribution is the structural equalisation of estimated preferences, access and earning opportunities followed by recomputation of money-metric welfare inequality. The backup slide adds the work on choice-set representation and restricted choices that matters most for separating preferences from opportunities.

$j = (e, k, h, w)$ 4 occupation groups $C_i(j)$ priced through EUROMOD

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

L_i
value of leisure: age; children for
women

β_c
weight on log consumption,
estimated

λ_c
units convention

In words: a household's systematic value of a job is the value of the leisure it leaves plus the log of the consumption it pays.

A job is a package: an employment state, an occupation group, weekly hours and an hourly wage. Every package is run through the French tax-benefit system, so consumption is the household's actual disposable income at that job. A couple chooses a pair of packages under a shared budget. The household-specific leisure coefficients are the preference channel; the child term is a reduced-form time-constraint shifter.

Opportunities: the choice probability

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

access
employment mass

hours
bands

occupation
by sex

wage offers
education, experience

valued **or** plentiful

In words: the probability of being seen in a job is its value times its availability, normalised over all jobs.

The opportunity density g says how intensely a market job is available before any choice; non-employment is the reference state. A job is chosen often when it is valued or when it is plentiful. If availability heterogeneity is omitted, some of its effects can be absorbed by the estimated utility component. Estimating the density household by household is what lets reachability show up as reachability.

Access, the employment mass of market jobs, moves with local unemployment exposure, region, urban or rural location, and year. Hours opportunities are banded, occupation availability is sex-specific, and wage offers are log-normal, located by education and experience.

How are preferences and opportunities separated?

Preferences	smooth systematic utility over hours
Access	local unemployment and geography, excluded from utility
Hours opportunities	band structure
Wage opportunities	education and experience shift the offer distributions
Budgets	EUROMOD generates heterogeneous nonlinear consumption schedules

These restrictions give different empirical variation to preferences and opportunities.

This is the identification strategy. Preferences are smooth in hours, so a spike in observed hours at a band is read as availability, not as a kink in tastes. Local unemployment exposure and geography shift access and are excluded from utility. Hours opportunities have a band structure. Education and experience shift the wage-offer distribution. And EUROMOD gives every household its own nonlinear budget, so the consumption value of the same job differs across households. Each block of the model is moved by different empirical variation. The deeper observational-equivalence discussion is in the backup.

EU-SILC France

collected 2016, incomes 2015

1,540 single-adult households

2,223 couple households

EUROMOD

French 2015 policy system

$h \in [5, 70]$ $w \in [2, 590]$

100 priced alternatives per household

French EU-SILC, with incomes referring to the year before collection; taxes, benefits and disposable income from EUROMOD under the policy system in force over the income period. The samples are single-adult and couple households of working age, not in education and not receiving a pension, estimated separately. Each household’s likelihood uses its observed job plus sampled alternatives, each run through the tax-benefit model.

Household-specific opportunities

free coordinates 41

population fit 0.0140

Common opportunities

free coordinates 10 / 16

population fit 0.0273 / 0.0264

criterion worse by at least 141.64

choices reflect opportunities, not preferences alone

single adults; population fit is the mean absolute deviation of predicted from observed moments

The first subquestion. Take away the household-specific opportunity block, re-estimate everything else, and the model does worse on the same households, the same sampled alternatives and the same criterion. Population fit roughly doubles its error, concentrated where the opportunity block does its work. Choices reflect opportunities, not preferences alone. Fit details, including the thirty-seven-hour mass point and the groups where accuracy is withheld, are in the backup.

Selected estimates

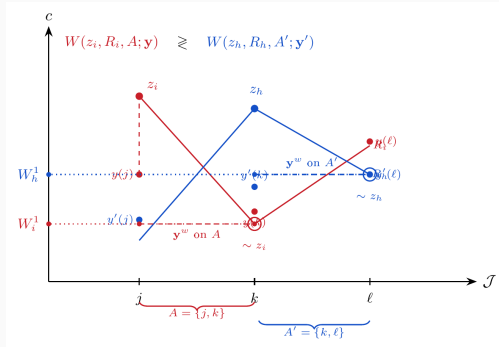
	Single adults	Couples
β_c consumption weight	2.0387 (0.2917)	2.1017 (0.2939)
$\beta_{E,s}$ local unemployment exposure, in access	−1.4422 (0.2356)	−1.1924 (0.1557)
β_{wH} high education, wage-offer location	0.1491 (0.0308)	0.1817 (0.0197)
σ log-wage dispersion	0.3815 (0.0133)	0.3631 (0.0072)

cluster-robust standard errors in parentheses; the two populations are estimated separately

Four estimates that carry the economics; the full vectors are in the backup.

The consumption weight is positive and similar across the two separately estimated models; it converts a non-consumption advantage into a proportional consumption adjustment in both welfare measures. Higher local unemployment exposure shifts employment opportunity mass downward for single adults and for couples. Higher education shifts the wage-offer distribution upward, and log-wage dispersion is common within each model. These are structural associations inside the estimated model.

Welfare: attained outcomes versus job prospects



Own-set equal-consumption equivalents: each individual gets a common consumption on every job in their own ability set; the level at which the preferred job is indifferent to the attained bundle is their money metric, comparable across individuals. Adapted from Haydar and Maniquet (2026), work in progress.

Estimated utility is not an interpersonal welfare index, because preferences differ. Each situation is translated into euros through an indifference condition with an explicit reference, and the choice of reference is the welfare question.

Both money metrics descend from this construction in the companion theory paper: give every job in a person's own set the same consumption, and find the level at which the preferred job is exactly as good as the attained bundle. The next two slides take the construction to the data in two ways: attained outcomes, and job prospects.

What constant consumption over the household's own job environment is equivalent to its prospect?

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

J_i
the actual prospect

H_i
same jobs, equal pay

M_i^{EA}
flat-consumption equivalent

In words: EA is the constant monthly consumption at which a prospect with the household's own jobs and availability, but equal pay everywhere, is worth exactly as much as the prospect it actually faces.

Every job enters J weighted by how available it is and how much it is valued; H keeps the same jobs, availability and preferences with pay differences removed. Opportunities enter welfare directly: access moves which jobs carry weight, and earning opportunities move what they pay. I show this perspective first because the talk is about opportunities.

What is the money equivalent of the attained bundle?

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

o

non-employment, reachable by all

j_i^{obs}

the only route for opportunities

In words: ATT is observed consumption, adjusted by the leisure the job costs relative to not working, converted at the rate one over the consumption weight.

The reference is non-employment, which every household can reach; on the estimated domain it is also the equal-pay preferred job for every household, so the measure is the staying-home equivalent. Opportunities matter through which job is attained and what it pays.

The two measures answer different welfare questions: attained outcomes versus job prospects. Neither corrects the other.

Inequality and Shapley decomposition

P	A	B	held fixed
preferences: age, children	local unemployment exposure, region, urban or rural location, year	wage-offer location	household resources, needs and composition

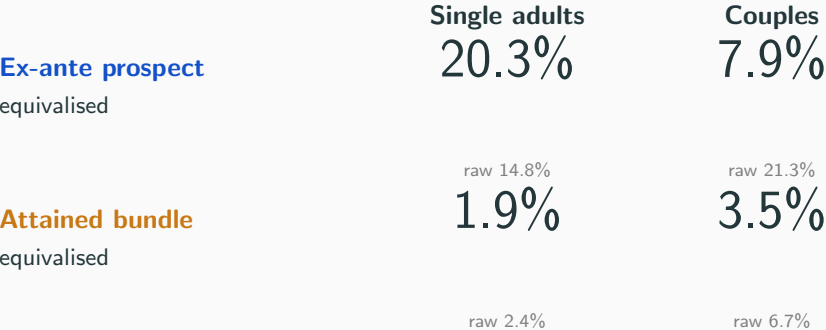
$$\phi_k^P = \sum_{S \subseteq \{P,A,B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[v^P(S \cup \{k\}) - v^P(S) \right]$$

$$v^P(S) = I^P(\emptyset) - I^P(S) \quad I^P: \text{household-weighted Gini of } M^P$$

In words: a channel's Shapley contribution is its marginal effect on inequality, averaged over every order in which the three equalisations can be introduced; the three contributions add up exactly to the total change.

P equalises age profiles and the child-related shifter. A equalises local access: local unemployment exposure, region, urban or rural location, and year. B equalises the systematic wage-offer location. Household resources, needs and composition are held fixed in every coalition. Each coalition is a full structural counterfactual: coefficients never change, welfare is recomputed for every household under each perspective, and inequality is measured again.

How much is associated with opportunities?



A + B as % of the relevant baseline Gini

The two measures also differ in aggregation: the prospect measure inverts within each household; the attained-bundle measure averages inequality across simulated realisations.

The headline. Each number is local access plus earning opportunities as a percentage of that perspective's own baseline Gini, on the equivalised scale.

Under prospect welfare, the two opportunity channels account for 20.3 per cent of baseline welfare inequality for single adults and 7.9 per cent for couples. Under attained-bundle welfare they account for 1.9 and 3.5 per cent. The small grey numbers are the raw household scale.

The ATT calculation is the less numerically mature of the two and is a useful benchmark rather than the sole basis of the paper.

The two measures also differ in aggregation. The reversal is consistent with the two perspectives valuing different objects, but the aggregation difference is not controlled for, and separating the two is ongoing work.

Which opportunity channel matters?



prospect welfare, equivalised: % of baseline Gini

single adults: access > earnings couples: earnings > access

attained-bundle welfare: earnings > access for both populations

Now split the prospect-welfare number into its two channels, each as a percentage of the baseline Gini on the equivalised scale.

Under prospect welfare, access contributes more than earning opportunities for single adults; for couples, earning opportunities contribute more than access. Under attained-bundle welfare the ordering for single adults is reversed: earning opportunities are larger than access for both populations. The full tables, with Gini points and the raw scale, are one click away.

Where does baseline inequality go?

shares of baseline Gini · prospect welfare, equivalised

	Single adults	Couples
Preferences ϕ_P/l_0	0.1%	−1.8%
Local access ϕ_A/l_0	15.5%	3.7%
Earning opportunities ϕ_B/l_0	4.8%	4.2%
Other / outside current P-A-B decomposition $1 - (\phi_P + \phi_A + \phi_B)/l_0$	79.6%	93.9%
Total	100.0%	100.0%

Couples: A + B alone = 7.9%; P + A + B = 6.1% because the preference contribution is negative (−1.8%).

Current decomposition equalises P, A and B only.

The same prospect-welfare decomposition, laid out so that it visibly sums to the whole baseline Gini on the equivalised scale.

Under the current ex-ante decomposition, labour-market opportunities account for about 20.3 per cent of baseline welfare inequality for single adults and about 7.9 per cent for couples.

For couples the preference contribution is negative, so preferences plus access plus earnings is smaller than access plus earnings alone. The last row is everything the current decomposition does not equalise; what it contains is in the backup.

What is not claimed

Scope

Household resources, needs and composition
held fixed

Access: local access, not total opportunity

Preliminary decomposition

Not claimed

Not causal

No parameter uncertainty yet

Preferences are not responsibility

Neither perspective is designated primary

prospects versus attained outcomes: two different welfare questions

The scope, stated once.

The decomposition equalises preferences, local access and earning opportunities, and holds household resources, needs and composition fixed. Access is local access, not total opportunity. It is a structural accounting exercise, not a causal decomposition, and parameter uncertainty has not yet been propagated through it. The preference channel is not a responsibility channel: it contains a reduced-form child term that can capture constraints as well as tastes. And the two welfare perspectives answer different questions; choosing between them is a normative decision not made here.

1. Choices reflect opportunities, not preferences alone.
2. Prospects: access leads for single adults. Attained outcomes: earnings lead. Couples: earnings lead under both.
3. Opportunities account for a bounded part of welfare inequality; the preference contribution changes sign with equalisation.

Next: extend the decomposition to household resources and needs.

Three answers.

First, observed choices carry information about opportunities: a model that gives everyone the same jobs fits worse on the same households.

Second, the welfare question changes which labour-market inequality matters. Under prospect welfare, access leads for single adults; under attained-bundle welfare, earning opportunities lead; for couples, earning opportunities lead under both.

Third, local access and earning opportunities account for a bounded part of welfare inequality, and the preference contribution changes sign with equalisation.

Next: extend the decomposition to household resources and needs, after fixing the appropriate operator. Thank you.

Literature	EOp benchmarks	Literature		
Model	Utility specification	Opportunity density	Likelihood	Identification
	Observational equivalence			
Estimation	Preferences	Employment access	Hours and occupation	Wage offers
	Common opportunities	Inference	Parameter uncertainty	
Fit	Extensive margin	Fit by margin	Thirty-seven-hour point	Matched households
	Marginal rate of substitution			
Welfare	ATT derivation	EA derivation	EA validation	Equivalisation
	Prospect reference	ATT numerical design	Aggregation orders	
Decomposition	Central figure	EA Gini points	ATT Gini points	EA channel shares
	ATT channel shares	Composition within P/A/B	Shapley vs one-factor	Outside P-A-B
	Robustness			
Data	Sample selection	EUROMOD pricing	Numerical design	

	share	outcome	data	measure	source
Italy	19.5%	gross earnings	SHIW 1993–2000	MLD; parental education	Checchi & Peragine (2010), p. 445
Brazil, urban	10–37%	male hourly earnings	PNAD 1996	Theil; five circumstances	Bourguignon et al. (2007), abstract
France	12.9%	post-tax earnings	EU-SILC 2005	MLD; five circumstances	Brunori et al. (2013), Table 1

These benchmarks

shares of earnings inequality attributed to observed circumstances

This paper

a share of welfare inequality from estimated latent opportunity components, resources and needs held fixed

Full citations. Checchi and Peragine, Inequality of opportunity in Italy, Journal of Economic Inequality: the Italy share is in their decomposition table and text. Bourguignon, Ferreira and Menéndez, Inequality of opportunity in Brazil, Review of Income and Wealth: the range is stated in the abstract and conclusion. Brunori, Ferreira and Peragine, Inequality of opportunity, income inequality and economic mobility, IZA Discussion Paper: France is in their cross-country table, which reports the estimate of Checchi, Peragine and Serlenga; that primary study is not in our corpus.

Comparability. These estimates are lower-bound shares of earnings or income inequality attributed to observed circumstances: adding circumstances can only raise them. This paper’s share is a share of welfare inequality associated with estimated latent opportunity components, with household resources and needs held fixed. The outcomes, the circumstance sets and the methods all differ, so the numbers orient the audience rather than rank countries or papers.

Choice sets and restrictions	Latent jobs and welfare	Inequality of opportunity and decomposition
Aaberge, Colombino & Wennemo (2009): alternative representations of the choice set	Aaberge, Dagsvik & Strøm; Dagsvik & Strøm; Dagsvik & Jia; Capéau, Decoster & Dekkers	Roemer; Checchi & Peragine; Bourguignon, Ferreira & Menéndez; Brunori, Ferreira & Peragine
Beffy, Blundell, Bozio, Laroque & Tô (2019): labour supply with restricted choices	Fleurbaey & Maniquet; Decoster & Haan; Bargain et al.; Jacquet, Jia & Thoresen	Shorrocks; Sastre & Trannoy; Creedy & Hérault; Mühlhan

Two papers matter most for separating preferences from opportunities. Aaberge, Colombino and Wennemo compare alternative representations of the choice set in labour-supply models, which is exactly the object our opportunity density replaces with an estimated one. Beffy, Blundell, Bozio, Laroque and Tô model labour supply with restricted choices, the same concern that hours opportunities are not freely chosen. The other columns list the latent-jobs, welfare, inequality-of-opportunity and decomposition references behind the main slides.

Without restrictions

a taste against certain hours and a scarcity of jobs at those hours predict the same choices

With restrictions

smooth tastes, banded availability and excluded access shifters move different parts of the choice distribution

If availability heterogeneity is omitted, some of its effects can be absorbed by the estimated utility component.

maintained restrictions, not tested · regional variation is not a causal design

Choices alone do not separate a taste for leisure from a scarcity of jobs at that number of hours: the two can be observationally equivalent. The separation comes from maintained functional-form and exclusion restrictions: smooth preferences against banded hours opportunities, access shifters excluded from utility, and wage offers independent of hours given occupation. If availability heterogeneity is omitted, some of its effects can be absorbed by the estimated utility component. These restrictions are maintained rather than tested, and the regional variation supports the access block only conditional on them.

Backup: attained-bundle numerical design

Sample	Counterfactuals	Checks
the already-priced estimation panel; no re-estimation and no new pricing	equalised covariates set to household-weighted means; attained bundle simulated by Gumbel-max with common random numbers	replications at fixed estimates; an independent second seed; exact Shapley closure
short weekly hours are sparsely represented in the integration panel		
Monte Carlo ranges are simulation spreads, not confidence intervals		
◀ Back ▶ Appendix map		

How the attained-bundle decomposition is computed. It reuses the already-priced estimation panel. In each coalition the household-constant covariates of the equalised channels are replaced by household-weighted means, and each household's attained bundle is simulated with the Gumbel-max rule under common random numbers, so realised taste draws carry through every coalition. The money metric and the Gini are recomputed, and each coalition value averages many replications at fixed estimates. An independent second seed reproduces the allocation, which closes exactly. The main open issue is that the panel represents short weekly hours sparsely; the effect of that on the attribution has not been fully quantified, which is why this calculation is still being validated.

Backup: two aggregation orders

Attained bundle

Gini within each simulated realisation, then averaged

$$I^{\text{att}}(S) = \frac{1}{R} \sum_r \mathcal{G}_w \left(\{ M_{i,r}^{\text{att}}(T_S x_i) \}_i \right)$$

r : simulated realisation of attained bundles, common random numbers across coalitions

the average of realised inequality versus the inequality of household equivalents

Prospect

inverted within each household, then one Gini

$$I^{\text{EA}}(S) = \mathcal{G}_w \left(\{ M_i^{\text{EA}}(T_S x_i) \}_i \right)$$

$M_i^{\text{EA}} = \lambda_c \exp\{(\log J_i - \log H_i)/\beta_c\}$, integrated over the household's prospect

The two decompositions apply the same operators and the same Shapley rule, but they aggregate in different orders. The attained-bundle object simulates attained bundles, computes the household-weighted Gini within each simulated realisation, and averages those Ginis over realisations. The prospect object integrates over each household's prospect, inverts to a money metric within the household, and then takes a single household-weighted Gini. The ex-ante money metric is not the expectation of the attained-bundle money metric, so the average of inequality and the inequality of household equivalents are different objects. The reversal is consistent with the two perspectives valuing different objects, but the aggregation difference is not controlled for, and separating the two is ongoing work.

Backup: the prospect reference

$$M_i^{\text{EA}} = \left(\int C_i(j)^{\beta_c} \omega_i(j) d\nu(j) \right)^{1/\beta_c}, \quad \omega_i(j) = \frac{e^{L_i(j)} \widehat{g}_i(j)}{H_i}$$

Own-opportunity reference

the reference prospect keeps the household’s own jobs, availability and preferences, and equalises pay

Weighted generalised mean

the metric is a generalised mean of consumption with exponent β_c , weighted by leisure value and availability

the reference moves with the household’s opportunity environment

Two readings of the ex-ante metric. First, its reference is the household’s own opportunity environment with pay equalised, so the reference itself moves when opportunities change: equalising access changes both J and H. Second, dividing J by H shows that the metric is a weighted generalised mean of consumption across jobs, with exponent the consumption weight and weights proportional to the leisure value of each job times its availability. A common rescaling of the density cancels from those weights. Whether the metric is monotone in opportunity expansions is a separate question, not discussed in the main talk.

Estimated	Not yet propagated	Simulation checks
cluster-robust standard errors for every estimated coordinate, clustered on the household	estimation uncertainty has not been carried through either welfare decomposition	second seeds and repeat runs measure numerical noise, not sampling uncertainty
the decomposition shares are point estimates without confidence intervals		

What is and is not known about uncertainty. Every estimated coordinate has a cluster-robust standard error. Estimation uncertainty has not been propagated through either decomposition, so the shares on the result slides are point estimates without confidence intervals, and differences between them are not statistically established. The seed and repeat-run comparisons measure numerical noise in the calculation, not sampling uncertainty. Propagating parameter uncertainty is a next step.

Backup: utility specification

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right), \quad L_i(j) = \beta_\ell^g(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_i(j); \theta_\ell^g)$$

$$\tilde{\ell}_i(j) = \frac{80 - h}{10}, \quad \mathcal{B}(z; \theta) = \frac{z^\theta - 1}{\theta}$$

$$\beta_\ell^g(\mathbf{x}_i)$$

intercept, centred age and its
square (units of 10 years)

$$k_i$$

children under 20, women only

couples

sum of the two spouses' leisure
terms

The full systematic utility. Leisure is weekly time left from a maintained endowment, rescaled, and passed through a Box-Cox transform with a group-specific curvature. The leisure weight varies with centred age and, for women, with the number of children; that child term is a reduced-form time-constraint shifter, not pure taste. The consumption weight is estimated. The shock is extreme value with unit scale.

Backup: opportunity density

$$g_i(o) = 1, \quad g_i(j) = g_i^E \, g_i^H(h) \, g_i^{\text{Occ}}(k) \, g_i^W(w \mid k) \quad \text{for market jobs } j \neq o$$

access

g_i^E : local unemployment exposure, region, urban or rural location, year

hours

g_i^H : elevated bands over a residual set

occupation

g_i^{Occ} : sex-specific group masses

wage offers

g_i^W : log-normal, truncated to [2, 590]

wage-offer location $\mu_i(k)$: education, potential experience and its square, occupation

Non-employment is the reference state with density one. Every market job carries access, the employment mass, times hours, occupation and wage-offer factors; for couples, each employed spouse's package carries its own factors. This is the form of the executed criterion, where the access index multiplies the working indicator. The unemployment exposure enters access scaled by 10. The wage location depends on education, potential experience in units of 20 years and occupation, with a common dispersion. The hours density is a density elevation over intervals, not an atom at a particular hour.

Backup: sampled-alternative likelihood

$$V_{ij} = v_i(j) + \log g_i(j) - \log q_{ij}$$

$$\Pr(y \mid \mathcal{C}_i) = \frac{n_y \exp V_{iy}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}$$

q_{ij}
proposal density, fitted out of fold

$\mathcal{C}_i$
observed job and 100 draws: 101 rows

n_y
multiplicity of a repeated draw

The choice set is a continuum, so the likelihood is evaluated over sampled alternatives drawn from a proposal density and corrected by it. The proposal is a computational device: fitted out of fold so a household's own outcome never enters the proposal used to score it, and it enters no welfare measure. For couples the proposal draws the joint participation regime first.

Backup: identification

Choices alone do not separate a taste for leisure from a scarcity of jobs at that number of hours. Three restrictions do the work: a spike at a band of hours is read as availability rather than a kink in tastes; excluded shifters enter availability and not preferences; and wage offers are independent of hours given occupation. These are the restrictions of the latent-jobs literature, maintained here rather than tested. No causal effect of geography is claimed.

Functional form

smooth preferences in hours;
banded hours opportunities

Exclusion

local unemployment exposure and
geography shift access, not utility

Independence

wage offers independent of hours,
given occupation

maintained, not tested · not a causal design

Backup: preference parameters

The preference block, with the parameter identifiers used in estimation. The consumption weight is common within each population. One single-adult coordinate, the female age-square term, sits at a box endpoint, so its standard error is not reported and it is excluded from the interior curvature. Levels are not comparable across the two separately estimated models.

	Single men	Single women	Couple men	Couple women
leisure weight, intercept	8.5199 (3.5480)	5.8683 (2.1346)	3.9814 (0.7466)	13.1029 (4.7760)
	beta_l0_sm	beta_l0_sf	beta_l0_m	beta_l0_f
leisure weight, age	1.4820 (1.0236)	0.0678 (0.4792)	−0.0080 (0.0287)	−0.1228 (0.1260)
	beta_l_age_sm	beta_l_age_sf	beta_l_age_m	beta_l_age_f
leisure weight, age squared	0.7783 (0.7732)	1.0000 (at bound)	0.0065 (0.0030)	0.0073 (0.0103)
	beta_l_age2_sm	beta_l_age2_sf	beta_l_age2_m	beta_l_age2_f
leisure weight, children	—	0.1666 (0.4422)	—	−0.3055 (1.1472)
		beta_l_nkids_sf		beta_l_nkids_f
leisure curvature θ_ℓ	−1.6263 (0.3301)	−0.9274 (0.2161)	−0.9761 (0.1357)	−1.6863 (0.2458)
	theta_l_sm	theta_l_sf	theta_l_m	theta_l_f
consumption weight β_c	2.0387 (0.2917)		2.1017 (0.2939)	
	beta_c		beta_c	

cluster-robust standard errors; parameter identifiers in grey; the male children term in couples is structurally zero

Backup: employment-access parameters

	identifier	Singles	Couples
employment constant	beta_E; beta_E_m, beta_E_f	−3.1735 (0.4053)	men −2.1233 (0.3195) women −2.8148 (0.3021)
local unemployment exposure	beta_E_gsur	−1.4422 (0.2356)	−1.1924 (0.1557)
densely populated	beta_E_drgur	−0.0288 (0.2209)	−0.1823 (0.1657)
intermediate density	beta_E_drgmd	0.0641 (0.2614)	−0.4174 (0.1861)
Region 2	beta_E_drgn2	−0.3780 (0.3292)	−0.1546 (0.2442)
Region 3	beta_E_drgn3	−0.1117 (0.3891)	0.0873 (0.2808)
Region 4	beta_E_drgn4	−0.8218 (0.3810)	0.0150 (0.3068)
Region 5	beta_E_drgn5	−0.5162 (0.3274)	−0.1455 (0.2555)
Region 6	beta_E_drgn6	−0.7222 (0.3528)	−0.2844 (0.2748)
Region 7	beta_E_drgn7	−0.5374 (0.3503)	−0.1396 (0.2651)
Region 8	beta_E_drgn8	−0.4512 (0.3401)	−0.1659 (0.2660)

references: the first region and thinly populated areas; couples share the shifters across spouses

The employment-access block of market jobs, with identifiers. The access index multiplies the working indicator, so non-employment is the reference. Local unemployment exposure lowers employment opportunity mass in both populations. Regional and urbanisation indicators shift access conditional on the exclusion restriction.

Backup: hours and occupation opportunities

Hours bands			
	Singles	Couple men	Couple women
part-time lower	0.0298	−1.1094	−0.5173
	beta_h_pt1	beta_h_pt1_m	beta_h_pt1_f
part-time upper	0.7080	0.0835	1.1586
	beta_h_pt2	beta_h_pt2_m	beta_h_pt2_f
narrow full-time	2.0656	2.2888	2.0284
	beta_h_f35	beta_h_f35_m	beta_h_f35_f
full-time upper	1.9387	2.3801	1.6825
	beta_h_ft	beta_h_ft_m	beta_h_ft_f
long hours	−0.0834	0.6977	−0.3109
	beta_h_lh	beta_h_lh_m	beta_h_lh_f

Occupation masses				
	Singles		Couples	
	men	women	men	women
Occupation 2	−1.3737	0.1135	−1.5097	0.2074
	beta_occ_2_m	beta_occ_2_f	beta_occ_2_m	beta_occ_2_f
Occupation 3	−2.1096	−0.4060	−2.2532	−0.1804
	beta_occ_3_m	beta_occ_3_f	beta_occ_3_m	beta_occ_3_f
Occupation 4	−0.3259	0.5632	0.1594	0.8159
	beta_occ_4_m	beta_occ_4_f	beta_occ_4_m	beta_occ_4_f

estimates with identifiers; standard errors in the paper

Hours opportunities are elevated densities over five bands relative to a residual set; hours coefficients are common across single men and women and spouse-specific for couples. Occupation masses are sex-specific, relative to the craft, agricultural, operator and elementary group. Identifiers are shown under each estimate.

Backup: wage-offer equation

The wage-offer equation, with identifiers. The offered log wage is normal with this location and a common dispersion, truncated to the wage support and renormalised. The singles and couples models are estimated separately, so the two experience profiles are not restricted to agree.

	identifier	Singles	Couples
intercept	beta_w0	2.0138 (0.0561)	2.0480 (0.0335)
low education	beta_w_educL	0.0566 (0.0372)	−0.0433 (0.0217)
high education	beta_w_educH	0.1491 (0.0308)	0.1817 (0.0197)
potential experience	beta_w_pexp	0.2408 (0.0862)	0.5643 (0.0546)
experience squared	beta_w_pexp2	−0.0282 (0.0390)	−0.1598 (0.0243)
Occupation 2	delta_occ_2	−0.0347 (0.0399)	−0.0740 (0.0231)
Occupation 3	delta_occ_3	0.0622 (0.0383)	0.0394 (0.0228)
Occupation 4	delta_occ_4	0.2785 (0.0368)	0.2146 (0.0223)
log-wage dispersion σ	sigma	0.3815 (0.0133)	0.3631 (0.0072)

medium education is the reference; experience in units of 20 years

Backup: common versus household-specific opportunities

The two re-estimated common-opportunity benchmarks restrict the opportunity block and re-estimate everything else. Both are worse on the same criterion, and population fit worsens where the opportunity block does its work. A better criterion does not by itself establish that a mechanism has been identified.

Single adults	free coordinates	criterion	population fit
household-specific opportunities	41	6253.463	0.0140
common opportunity distribution	10	6403.974	0.0273
common shape; employment and hours in utility	16	6395.108	0.0264

same households, sampled alternatives and criterion; not converted into a formal test

Single adults	Couples
free coordinates 41, interior 40	free coordinates 47, interior 47
criterion 6253.463	criterion 10283.034
smallest Hessian eigenvalue 0.1669	smallest Hessian eigenvalue 0.0771
cluster-robust standard errors on the household	eigenvalues on the interior block

Curvature and inference. The exact Hessian on the interior block is positive definite for both models. Several terminal paths from different starts agree to numerical precision; that supports a stable local solution from the starts tested, not global uniqueness. One single-adult coordinate at a box endpoint is handled under the active-set convention.

Backup: extensive-margin fit

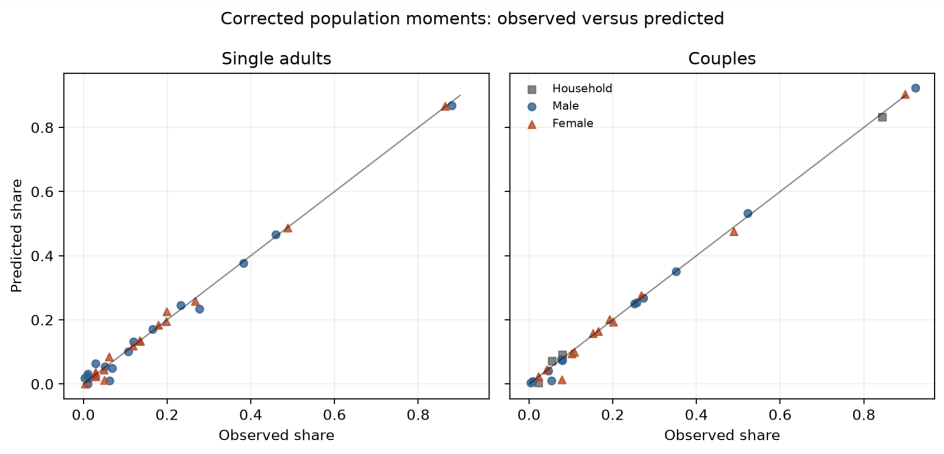
Extensive-margin accuracy is reportable for women in each sample and withheld for men, because numerical integration error is too large relative to sampling variation. The integration ratio is that comparison; the rule was fixed before the evaluation.

	weighted accuracy	model-simulated range	integration ratio
single women	85.7%	80.0–85.9%	0.096
coupled women	89.8%	88.1–91.2%	0.049
single men	withheld	—	0.361
coupled men	withheld	—	0.353

withheld where numerical integration error is too large relative to sampling variation

Backup: fit by margin

Corrected population fit: observed against predicted shares, margin by margin. Participation and occupation margins are reproduced closely in aggregate; the hours fit is uneven. Population predictions integrate the model over opportunities and taste shocks, so this is not an in-sample fitted choice.



Backup: the thirty-seven-hour point

single men

5.2

pp

single women

3.8

pp

coupled men

4.4

pp

coupled women

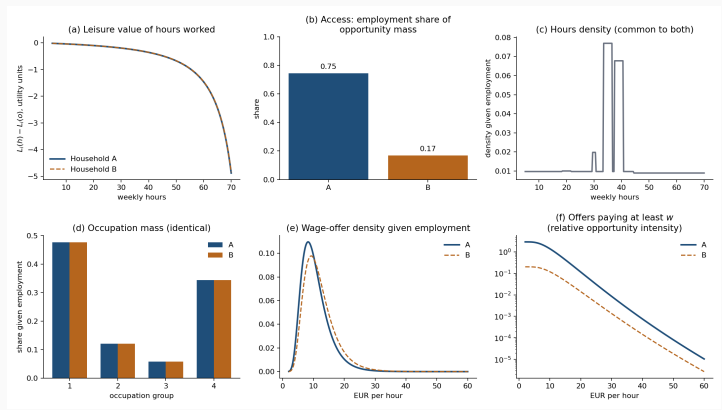
6.5

pp

observed minus predicted share at the observed thirty-seven-hour mass point

The common hours mismatch is underprediction of the observed concentration at exactly thirty-seven hours. It is a finding about that mass point, not about the model's neighbouring full-time range, and it is kept as its own result.

Backup: matched households

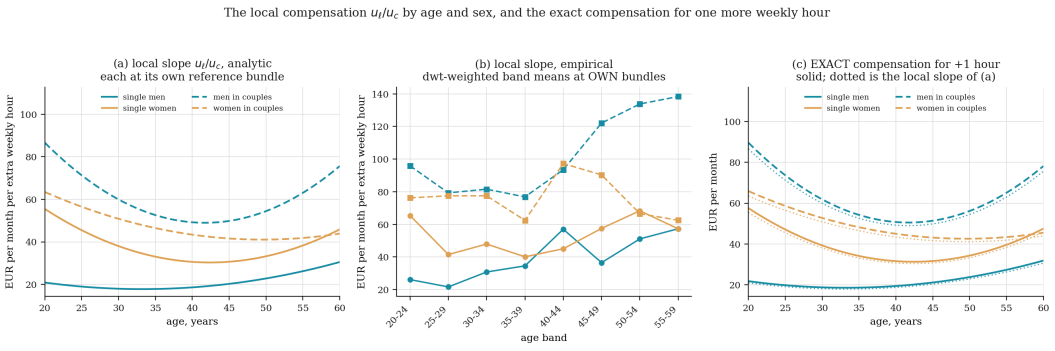


access: $A = 14.5 \times B$ wage-offer location: $B + 11.5$ log points

What unequal opportunity means inside the model. Two employed single men in the same occupation group, hours band and wage quintile, with nearly identical estimated leisure profiles, selected by a stated rule: a teaching example, not causal and not representative. Employment is 0.75 of A's opportunity mass and 0.17 of B's. B's wage offers are better, yet A faces more offers paying at least any given wage over essentially all offer mass. Neither is better placed on every margin.

Backup: marginal rate of substitution

The marginal rate of substitution between leisure and consumption, by age and sex, in euros per month per additional recurring weekly hour of leisure. It is a compensation along an indifference curve, not a behavioural response to a wage change. The coefficients are coordinates; this is the economics.



Backup: attained-bundle derivation

$$L_i(o) + \beta_c \log\left(\frac{M_i^{\text{att}}}{\lambda_c}\right) = L_i(j_i^{\text{obs}}) + \beta_c \log\left(\frac{C_i^{\text{obs}}}{\lambda_c}\right)$$

$$\implies M_i^{\text{att}} = C_i^{\text{obs}} \exp\left\{\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}\right\}$$

Reference job

the preferred job when every job pays the same:
here non-employment for every household

Staying-home equivalent

a verified property of this specification, not a
general theorem

The attained-bundle money metric solves an indifference condition between the observed job and consumption and the reference state at the unknown consumption. The normaliser cancels. The general construction uses the job the household would most prefer at equal pay; on the current domain non-employment maximises the non-consumption index for every household, so the measure is the staying-home equivalent. With a different leisure specification the two could separate.

Backup: ex-ante derivation

$$J_i = \int_{\Omega_i} e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(m) = \left(\frac{m}{\lambda_c} \right)^{\beta_c} H_i, \quad H_i = \int_{\Omega_i} e^{L_i(j)} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(M_i^{\text{EA}}) = J_i \implies M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

a common rescaling of $\widehat{g}_i$ cancels $\cdot \log J_i$: expected utility of the best reachable job, up to a constant

The reference prospect keeps the household's jobs, availability and preferences and pays the same consumption at every job; its value is the power of that consumption times H. Setting it equal to the actual prospect and solving gives the ex-ante money metric. Under the extreme-value shocks, log J is the expected utility of the best reachable job up to a constant common to J and H, which is why taste-shock variety counts as welfare-relevant: a normative position.

Backup: ex-ante numerical validation

Identities

constant-consumption identity

one-state identity

common intensity-rescaling invariance

flat-reference inversion

exact Shapley adding-up

Numerics

exact reference integral on the priced se

effective sample size and weight concentration

two-seed stability of welfare, Gini and Shapley

finite positive welfare for every household

independent reimplementations

all passed for both populations · a correct computation, not causal identification or parameter uncertainty

The ex-ante calculation passed every declared check for both populations, including an independent reimplementations, with the integration design fixed before inspecting any welfare level or Shapley value. That certifies the computation given the model and operators. It does not establish causal identification, it does not quantify statistical uncertainty in the estimated parameters, and it does not make either welfare perspective normatively correct. Channel orderings are unchanged across the reported reference-domain sensitivities.

Backup: equivalisation

$$M_i^{p,eq} = \frac{M_i^p}{e_i}, \quad e_i : \text{ modified-OECD scale}$$

Couples, ex ante: A + B

21.3% $\longrightarrow$ 7.9%

raw $\rightarrow$ equivalised; % of baseline Gini

Couples, ex ante: preferences

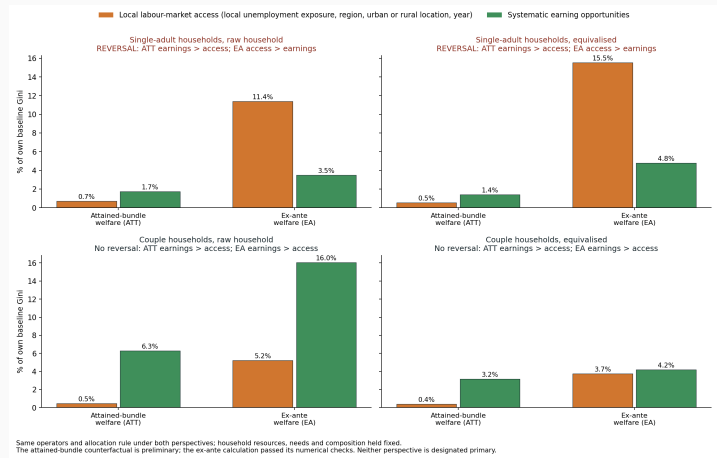
0.00291 $\longrightarrow$ -0.00212

Gini points, raw $\rightarrow$ equivalised

a normative convention, reported both ways; populations never combined or compared

Equivalisation is a normative convention and it is consequential for couples under the ex-ante perspective: it moves access plus earnings as a share of baseline inequality and turns the preference contribution negative. A negative Shapley term means that, averaged over orders, equalising preferences raises inequality; it does not show that preference heterogeneity is equalising in any causal or welfare sense. That is why both scales are always reported.

Backup: central figure



The published central figure: access and earning-opportunity contributions, each as a percentage of its own perspective's baseline Gini, under attained-bundle and ex-ante welfare, raw and equivalised. For single adults earning opportunities are larger under the attained bundle and access is larger under the prospect; couples show no reversal. Resources, needs and composition are held fixed.

Backup: ex-ante decomposition in Gini points

The full ex-ante table. Contributions are signed Gini points; access plus earnings is reported relative to each baseline Gini. The calculation passed its numerical checks, including an independent implementation. For couples the preference contribution changes sign with equalisation.

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.1595	0.00157	0.01812	0.00555	14.8	access > earnings
	equalised	0.1406	0.00020	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.1277	0.00291	0.00667	0.02049	21.3	earnings > access
	equalised	0.1200	−0.00212	0.00449	0.00504	7.9	earnings > access

exact Shapley contributions in Gini points; household-weighted

Backup: attained-bundle decomposition in Gini points

The attained-bundle table. Earning opportunities are larger than local access for both household types and both scales. The preference contribution changes sign with equivalisation in both samples, so no directional claim is made about it. The lower part of the hours range is sparsely represented in the integration sample, which is why this exercise stays preliminary.

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.2337	0.0012	0.0016	0.0040	2.4	earnings > access
	equivalised	0.2451	−0.0003	0.0013	0.0034	1.9	earnings > access
Couples	raw	0.2035	0.0064	0.0009	0.0128	6.7	earnings > access
	equivalised	0.1971	−0.0019	0.0007	0.0062	3.5	earnings > access

exact Shapley contributions in Gini points; preliminary; household-weighted

Backup: ex-ante channels, % of baseline

Each ex-ante Shapley contribution, in Gini points and as a percentage of its own baseline Gini, raw and equivalised. The negative couples preference term is kept. P, A and B do not exhaust baseline inequality: household resources, needs and composition and other dimensions are held fixed or left unallocated.

		P				A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0	
Single adults	raw	0.1595	0.00157	0.99	0.01812	11.37	0.00555	3.48	
	equivalised	0.1406	0.00020	0.14	0.02183	15.52	0.00673	4.79	
Couples	raw	0.1277	0.00291	2.28	0.00667	5.22	0.02049	16.04	
	equivalised	0.1200	−0.00212	−1.76	0.00449	3.75	0.00504	4.20	

each contribution divided by its own baseline Gini I_0 ; signed; the rows are not forced to sum to one hundred, because the remainder is outside the current P-A-B decomposition

Backup: attained-bundle channels, % of baseline

		P			A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0
Single adults	raw	0.2337	0.0012	0.53	0.0016	0.69	0.0040	1.73
	equivalised	0.2451	−0.0003	−0.13	0.0013	0.54	0.0034	1.40
Couples	raw	0.2035	0.0064	3.16	0.0009	0.45	0.0128	6.28
	equivalised	0.1971	−0.0019	−0.98	0.0007	0.37	0.0062	3.15

each contribution divided by its own baseline Gini I_0 ; signed; preliminary

The same layout for the attained-bundle perspective. Earning opportunities exceed access in every row; the preference contribution changes sign with equivalisation. These percentages are shares of each baseline Gini and are not forced to sum to one hundred.

Backup: composition of the explained component

$$s_k = \frac{\phi_k}{\phi_P + \phi_A + \phi_B}, \quad k \in \{P, A, B\}$$

ex-ante prospect, equivalised	Single adults	Couples
Preferences s_P	0.7%	−28.5%
Local access s_A	75.9%	60.6%
Earning opportunities s_B	23.4%	67.9%
Total	100.0%	100.0%

These are shares of the currently explained P/A/B change, not shares of total inequality. Negative components are possible under Shapley attribution.

These are shares of the currently explained P/A/B change, not shares of total inequality. Negative components are possible under Shapley attribution.

Within that currently explained component, most of the single-adult opportunity effect comes through access rather than earning opportunities. For couples the negative preference share means the access and earning shares together exceed the whole explained change. Only the equivalised scale is shown: on the raw scale the one-decimal couples shares would not close exactly.

Backup: Shapley versus one-factor equalisation

one-factor: $v^P(\{k\}) = I^P(\emptyset) - I^P(\{k\})$

Shapley: $\phi_k^P = \text{average over all orders of } v^P(S \cup \{k\}) - v^P(S)$

$$\phi_P^P + \phi_A^P + \phi_B^P = I^P(\emptyset) - I^P(\{P, A, B\})$$

one-factor effects need not add up · Shapley contributions do, exactly

A one-factor effect equalises one channel from the actual situation. Because the channels interact, those effects need not add up, and the answer would depend on the order chosen. The Shapley contribution averages each channel's marginal effect over every order and adds up exactly to the total change. This is the standard rule from the distributional decomposition literature; no new principle is claimed. Coalition-by-coalition Gini tables are not reproduced in these slides.

Backup: outside the current P-A-B decomposition

Held fixed

household resources, needs and composition

Retained

sex-specific and household-type-specific preference and opportunity blocks

Not equalised

wage-draw luck and the common offer spread

wage-offer location differences: at most about 0.10 log points

common offer spread: about 0.37 log points

next: extend the decomposition to household resources and needs, after fixing the appropriate operator

The last row of the percentage display is everything the current decomposition does not equalise. It is not a channel. It holds household resources, needs and composition fixed, it keeps the sex-specific and household-type-specific blocks, and it excludes wage-draw luck and the common offer spread. The location differences that the earning operator equalises are small next to that common spread, so the operator removes only a small part of earnings dispersion by construction. Extending the decomposition to household resources and needs requires fixing the appropriate operator first, because some covariates enter both preferences and needs; until then there is no fourth channel to allocate.

Backup: robustness of the attained-bundle decomposition

Second simulation seed

every coalition Gini and contribution
reproduces closely, including the preference
sign change

Observed job in the simulation

attained under every coalition by 2.9% of single
adults and 3.9% of couples
excluding it moves the total change by at most
9.3%, with no sign change

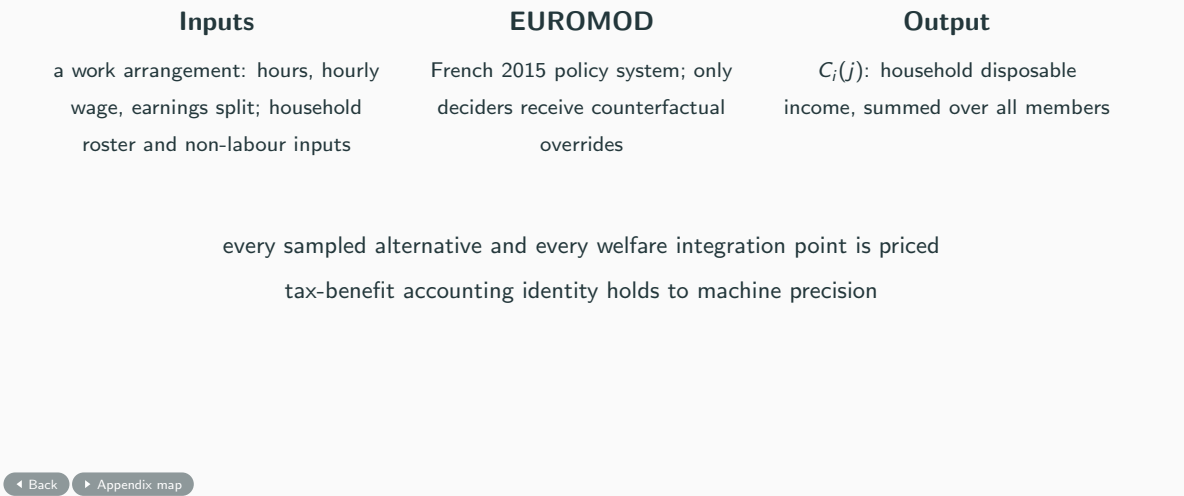
simulation checks, not confidence intervals

Robustness of the attained-bundle calculation. An independent second seed reproduces every coalition Gini and Shapley contribution closely. The simulation always includes each household’s own observed job; excluding it moves the total change by at most the amount shown, in the couples equivalised case, with no sign change. These checks validate the calculation for the stated game, not its maintained assumptions, and they are not confidence intervals.

Backup: sample selection

Unweighted households remaining after each screen. The largest exclusion is structural: multi-generational households, flat-shares and same-sex couples are outside the estimated population, and no result extends to them. Students and pension recipients are removed because the model does not define an offer set for them; the remaining screens are support restrictions.

screen	single-adult	couple
the raw France 2016 input file	4,038	5,965
one- or two-adult households (composition screen)	4,038	5,965
every adult aged 20 to 60	2,131	3,662
no adult still in education	2,036	3,521
no old-age, disability or survivor pension receipt	1,755	3,218
labour status in scope	1,598	2,412
no other employable or earning member	1,564	2,323
hours and wage inside the modelled support	1,555	2,275
hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider	1,543	2,223
observed chosen alternative priced at non-positive disposable consumption	1,540	2,223
estimation sample	1,540	2,223



For a given work arrangement the tax-benefit model receives the implied gross labour inputs together with the household's non-labour inputs and roster, and returns disposable income. Only deciders receive counterfactual overrides; other members keep their baseline values. The accounting identity linking original income, benefits, taxes and contributions to disposable income holds to machine precision at the person and household-alternative level.

Estimation

101

rows per household: the observed
job and 100 sampled alternatives

Predictive fit

2,048

common integration points per
household

Supports

[5, 70]

weekly hours; wages in [2, 590] per
hour

sampling devices are computational: they are not offers and enter no welfare measure

The numerical design. Estimation uses the observed job plus sampled alternatives per household. The predictive fit integrates over a separate, larger common set of priced integration points per household. The supports are the intervals on which the opportunity densities are defined. The ex-ante welfare integrals use their own certified design, chosen before any welfare result was inspected.